

Proposed Variables

Var No	Variable	Explanation	Comments
0	Status	Simple binary outcome: Signed or Not Signed	
1	Property class	Is it a class a or b building	Dropped. Quadrant categories could probably work better as indicators of value proposition.
2	Existing deals with partner	Count of finalized deals with partner at the time of Proforma creation	Accepted.
3	Expansion	Is this a new deal, or an expansion	Dropped. It's more exhausting to step back and ask whether it's a new partner or not.
4	Property category	Quadrants: do they have low occupancy and would we perform well	Accepted.
5	Deal type	Standard or Lease Up	Dropped. Not enough lease ups to justify wasting a variable in this.
6	Unit breakdown	Count of unit mix	Approved. Could use the core unit mix demand (ratio of Studios and 3-BRs to 1-BRs and 2-BRs)
7	Minimum term required	The lowest term length the IC allowed the salesperson to offer (e.g., 1, 3, or 5).	Approved.
8	Blended rent	Weighted average rent	Dropped. This is not a proposition, it's data mined from the partner's website.
9	Estimated occupancy rents		Dropped. This is accounted for inside the Multivariate Family Equivalent value proposition.
10	Revenue share ratio		Accepted.
11	Multifamily equivalent percentage		Accepted.
12	Net operating income (anual)	Logarithmic scale $\ln(\text{NOI})$	Approved. We need to test its Variance Inflator Factor (VIF). We need a measure of impact, not just ratios, which is why this is included. Among the variables that create NOI, the only one to be included is revenue share.

13	Salesperson ID		Dropped from the logit regression, included in the forest model. Too many dummies, it would kill the model and consume the events-per-variable (EPV) threshold.
14	Redline volume	Count of redline exchanges	The count of clauses contested
15	Redline friction	Ratio of pushback against redlines	A calculated ratio measuring pushback between the summatory of denied redlines between the summatory of requested redlines. If Redline volume = 0, then Redline friction = 0
16	Escalation	Count of approvals required by an executive	Approved
17	Time talking	Ratio of time the salesperson talked over the partner	Approved.
18	Open time	The number of days between opportunity creation and <i>the exact date the prediction is run.</i>	Dropped. A "Signed" deal's Open Time is the duration of the negotiation. A "Not Signed" deal's is just the amount of time it took a salesperson to click close the deal in cleanup.
19	Last action time	The number of days between the last logged activity and <i>the exact date the prediction is run.</i>	Approved.
20	Last action to close date	The time between the last action and the close date	Dropped. This metric cannot be calculated for open pipeline deals in a live predictive environment. Colinear with dependent variable.
21	valid open time	a boolean variable that equals 1 for opportunities with positive Open Time and 0 for the remaining opportunities	Dropped. All other time variables were dropped.
22	Penetration Ratio	Ratio of Landing units against total units in the building	Approved.

As a note, most of these variables and more may be included in the Random Forest Model, but including them in the Logistic Regression would consume the events-per-variable (EPV) threshold. Given that the model occupies different scales between absolute integers, ratios, and logarithmic scales, it is advisable to standardize all continuous variables to z-scores for the Logit model. The Random forest model does not require this correction. If $\ln(\text{NOI})$ exhibits high collinearity with the Penetration ratio and revenue share (a standard VIF threshold of > 5.0 will be used first), it might have to be dropped from the Logit model.

Variables exhibiting a VIF > 5.0 will be dropped sequentially, beginning with the highest VIF.

Given that the CRM data is not consistently purged, it is best to drop the time variables from the initial logit regression. Any time metric associated to an unsigned deal will introduce noise. We propose, in addition to the logit and random forest models, a Kaplan-Meier survival curve on the deals to solve the CRM quality problem. With this timestamp, we can create a distribution of successful deals to establish a probability threshold T for when 95% of deals are signed. Deals that are open, but past that point, are treated as censored or lost. This way, the KM survival curve defines when an “Open” deal is actually a “0” (Not Signed) for the other two models.

When applying this threshold to the dataset, we will filter the data as follows:

- If a deal is “Open” and Deal Age is less than or equal to the threshold, it’s dropped from the dataset (we don’t know the outcome of an active deal).
- If a deal is “Open” and Deal Age is greater than the threshold, its status is overwritten to 0 (Not Signed).

By defining T at the 95% cumulative probability mark, we acknowledge that there will be a 5% error rate. Since 5% of successful take took longer than the threshold to sign, we will be getting rid of a small portion of deals that did get signed, but were slow. We accept this margin of error as a necessary trade-off to sanitize the CRM data.

We are aware that there needs to be a much more robust analysis of negotiation tactics. For now, we will keep the following variables, and will add two more variables, at least, for the logit model to take negotiations into account.

The final variables for the initial Logit model are:

Var No	Variable	Definition
0	Status	Simple binary outcome: "1" (Signed) or "0" (Not Signed).
1	Benefit to partner	A simple Boolean which determines whether the partner's current occupancy rate crosses the threshold determined by the IC to benefit from our services or not.
2	Existing deals with partner	Count of finalized deals with partner at the time of Proforma creation.
3	Unit breakdown	A ratio measuring the unit mix proposal according to our strongest Member's demand (ratio of Studios and 3-BRs to 1-BRs and 2-BRs).
4, 5, 6	Terms offered	A dummy variable (independent Boolean) that shows the term lengths the IC allowed the salesperson to offer (e.g., 1, 3, or 5), where each variable represents 1 Year, 3 Years and 5 Years.
7	Revenue share ratio	The initial revenue share proposed for the main pitch (according to the term).
8	Multifamily equivalent percentage	The compared performance between the equivalent multifamily occupancy NOI and Landing's estimated performance.
9	Net operating income (annual)	Logarithmic scale $\ln(\text{NOI})$. Test VIF.
10	Redline volume	Count of redline changes contested.
11	Redline friction	A calculated ratio measuring pushback between the summatory of denied redlines between the summatory of requested redlines. If Redline volume = 0, then Redline friction = 0.
12	Escalation	Count of approvals required by an executive.
13	Time talking	Ratio of time the salesperson talked over the partner.
14	Penetration Ratio	Ratio of Landing units against total units in the building.

Assuming these 14 variables and the academic standard of 10 events per variable, we would need 14 variables x 10 events = 140 Signed Deals to run this Logit regression model. This is the traditional standard for Maximum Likelihood Estimation (MLE). We can be confident that the model will not overfit and that the p-values are reliable.